

Cognitive Architectures for Robotics

Standalone Localization Benchmark for Humanoid Robot Indoor Operation

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Introduction and Research Problem

Core Challenges in Autonomy

Deploying mobile robots in dynamic, unstructured environments reveals critical vulnerabilities:

Navigation Frameworks: They frequently lack the deterministic lifecycle management and rigorous security protocols required for safe operation.

The Localization Paradox: Traditional Monte Carlo Localization (MCL) relies on particle filters that ironically fail with highly accurate sensors.

Extremely narrow true posterior distributions cause a “sampling mismatch”.

This leads to rapid particle depletion and localization failure.

Limits of Long-Term Operation

SLAM Memory Constraints: As the mapped environment grows, continuous loop-closure matching causes unmanageable processing delays.

Geometric Alignment: Registering 3-D shapes without pre-extracted features remains a computationally intensive hurdle.

Biological Inspiration: Parallel stochastic systems in nature (e.g., RNA polymerization) use localized backtracking to correct mismatch errors, offering a mathematical blueprint for robotic error-correction.

State of the Art

Current Limitations

A review of open-source tools reveals significant gaps for industrial-grade autonomy:

Navigation: Legacy solutions struggle to keep up with multi-core sensor fusion and strict state-machine governance.

Geometric Alignment: Iterative Closest Point (ICP) is the standard but requires careful initialization to avoid incorrect local minima.

Localization: Conventional MCL relies purely on forward-guessing (motion model), causing starvation in low-noise scenarios.

SLAM: Current mappers map indefinitely, inevitably exceeding hardware memory and processing capabilities.

Research Formulation



Objectives and Hypotheses

Primary Objective: Develop robust, modular, and computationally efficient architectures capable of real-time state estimation and secure navigation.

Core Hypotheses:

Task Orchestration: Behavior trees in ROS2 will provide secure execution.

Memory Management: Bounding “Working Memory” guarantees capped SLAM times.

Localization Robustness: Mixture MCL will eliminate degradation from accurate sensors.

3-D Registration: ICP will robustly register arbitrary shapes minimizing distances.

A multi-faceted methodology spanning architecture, sampling, and modeling:

System Architecture (Nav2): Utilizing Behavior Trees to break down navigation logic into modular, fault-tolerant servers.

Mapping Pipeline (RTAB-Map): Mimicking cognitive processes (Short-Term, Working, Long-Term Memory) to guarantee bounded real-time processing.

Probabilistic Sampling: Combining standard forward-guessing with a “backward” dual sampling approach to populate correct regions.

Stochastic Modeling: Applying Laplace transforms to calculate explicit error-correction and first-passage times.

Cognitive Architecture Implementation

A foundational spatial representation is constructed via RTAB-Map (RGB-D + 3D LiDAR), mimicking cognitive spatial encoding.

Long-Term Memory: The synthesized global graph is serialized into an immutable .db database for pure localization tasks.

Geometric Filtering: Due to hardware constraints, strict filtering is applied (ignoring objects below 0.02 m).

Trade-off: The robot becomes cognitively “blind” to very low-profile impediments, optimizing the map for real-time navigation.

Reactive Obstacle Avoidance: The Subconscious Layer

Cognitive architecture divides reasoning into conscious (global planning) and subconscious (local reactive avoidance).

Local Costmap: Updated in real-time by LiDAR; virtually expands obstacles via an `inflation_radius` corresponding to the robot's footprint.

Dynamic Window Approach (DWA): Acts as the subconscious layer when dynamic threats intersect the trajectory.

Rapid Response: Simulates and scores short-term velocity trajectories, allowing the robot to unconsciously swerve without recalculating the global route.

Benchmarking & Results

Uni-modal Vulnerability vs. Multi-modal Robustness

Stress tests revealed a critical divergence in localization strategies:

Uni-modal Failure (2D LiDAR / AMCL)

- Dynamic actors occlude rays, destroying static spatial context.
- Physical collisions cause kinematic discrepancies.
- Result: Covariance Explosion and irrecoverable loss of global localization.

Multi-modal Resilience (Sensor Fusion)

- Fusing 2D LiDAR with 3D RGB-D visual odometry (RTAB-Map).
- Visual sensors track elevated features unoccluded by actors.
- Result: Continuous pose anchoring, overriding discrepancies and preventing divergence.

Scenario 1 & 2: Nominal vs. High Noise

Scenario 1: Nominal Conditions

- Standard noise (stddev 0.02 m).
- Consistent pose estimation (Error < 0.05 m).

Scenario 2: Sensory Uncertainty

- **High Noise:** Poorer representation leads to frequent re-convergence.
- **Extreme Noise (0.5m):** Profoundly distorts the point cloud, starving AMCL.
- **Result:** Catastrophic loss of localization.

Traiettorie: Extreme noise - localization_results.csv

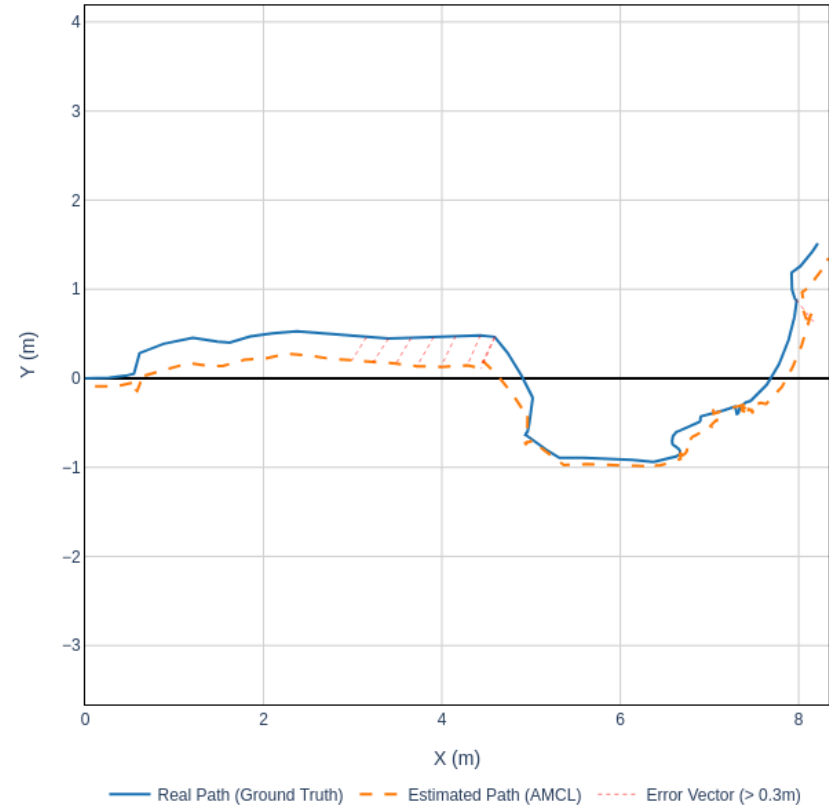


Figure 1: Extreme sensory noise starvation.

Scenario 3 & 4: Dynamics & Kidnapping

Scenario 3: Dynamic Obstacles

- Superior reactive DWA deviation.
- Limitation: Collisions occur in lateral/posterior blind spots (no 2D LiDAR coverage).

Scenario 4: Robot Kidnapping

- Abrupt teleportation causes immediate error explosion (> 10 m).
- Active “forced exploration” triggers fast recovery.
- **Result:** Asymptotic decay curve recovery.

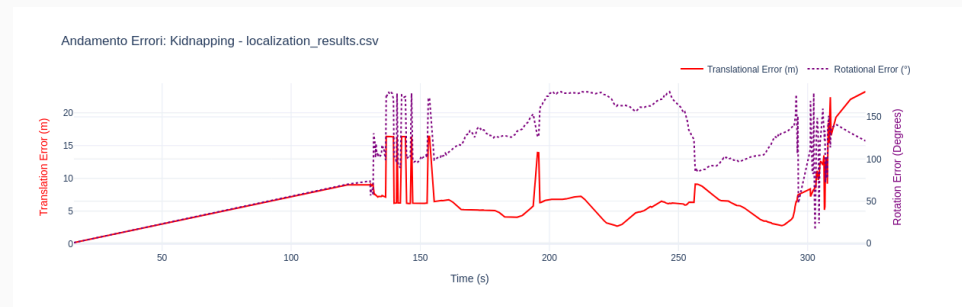


Figure 2: Error spike and asymptotic recovery.

Conclusions

- Strengths: Baseline framework is highly reliable. Capable of self-recovery from global position loss via active exploration.
- Vulnerabilities: Susceptible to severe sensor degradation and physical blind spots.

Final Recommendation: Safe deployment requires a **multimodal sensor fusion approach** to mitigate physical blind spots and noise susceptibility.

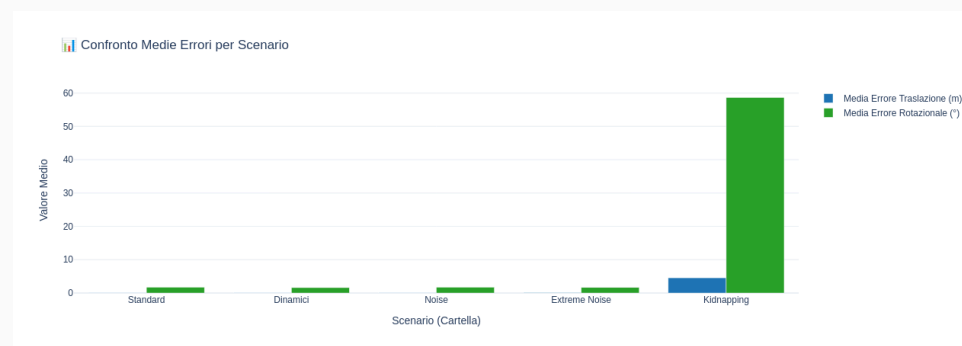


Figure 3: Mean errors across evaluated scenarios.